

High Ionic Conductivity in Poly(dimethyl Siloxane-co-Ethylene Oxide) Dissolving Lithium Perchlorate

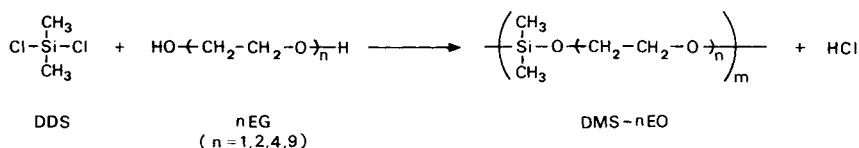
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High ionic conductivity in poly(ethylene oxide) (PEO)-alkali metal salt complexes has been the subject of a number of recent publications.¹⁻⁹ The conductivity reached 10^{-4} S/cm at moderate temperatures in some complexes.^{1,2} The complexation was considered to involve coordination of the cations to the ether oxygen atoms in the polymer backbone as was seen in the macrocyclic polyether complexes.¹⁻⁴ As a result the complexes were thought to have a helical structure.³⁻⁵ At the initial stage of the study the high ionic conductivity of the PEO complexes attributed to the intrahelical jumping process by analogy with fast ion transport in the inorganic solid electrolytes.¹ However, the evidence that ionic transport within the amorphous region was an important process in high ionic conductivity has been presented by several authors.⁶⁻⁹ From this point of view, high segmental mobility of PEO at ambient temperatures correlating to the low glass transition temperature ($T_g = -66^\circ\text{C}$) seems to contribute to the high mobility of the ionic carriers. The structural characteristics (i.e., polar ether oxygen exists at every three atoms in the polymer backbone) ensure cooperative interaction of the neighboring oxygen atoms and ions. This interaction seems to contribute to the generation of ionic carriers.

We reported previously that the studies on the viscous solutions formed by polyether oligomer and a salt gave much information about ion conductive behavior in polymers.^{10,11} Moreover, the conductivity of crosslinked films of the polyether dissolving a salt was found to be comparable to that observed in these solutions.¹² In this study, poly(dimethyl siloxane-co-ethylene oxide) (DMS-*n*EO) was synthesized in order to obtain ionic conductors with much higher conductivity than the PEO complexes. We suspected that the introduction of the dimethyl siloxane (DMS) units decreased the T_g 's of the copolymers without decreasing the solubility of salts.

EXPERIMENTAL

DMS-*n*EO was synthesized by means of a polycondensation reaction of dimethyldichlorosilane (DDS) with mono-, di-, tetra-, or nonaethylene glycol (*n*EG; *n* = 1, 2, 4, 9), according to the following reaction:



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Reagents and solvents were purified by conventional methods. Benzene solutions of the equimolar amounts of DDS and *n*EG were mixed and stirred under nitrogen atmosphere. The temperature of the contents was maintained at 10°C for the first 72 h, and at 60°C for the following 72 h. Finally, a dynamic vacuum was applied at 60°C for 48 h in order to complete this reaction and to remove the solvent and hydrogen chloride sufficiently. The DMS-*n*EO copolymers obtained were transparent and viscous liquids.

Anhydrous lithium perchlorate (LiClO_4) was dried under reduced pressure at 150°C for 16 h before use. LiClO_4 was dissolved in a benzene solution of DMS-*n*EO copolymer or PEO (molecular weight = 400). The solvent was evaporated completely under reduced pressure to obtain viscous electrolyte solutions. The concentration of LiClO_4 in the copolymers or PEO was represented by the molar ratio of LiClO_4 to ethylene oxide (EO) units ($[\text{LiClO}_4]/[\text{EO unit}]$).

Conductivity of the given electrolyte solutions was measured in a thermoregulated bath with a Toa-Electronics CM-20A digital conductance meter. Differential scanning calorimetry (DSC) was measured with a Rigaku Denki 8085 DSC apparatus at a heating rate of 20°C/min.

RESULTS AND DISCUSSION

In $^1\text{H-NMR}$ spectra of DMS-*n*EO copolymers, the signal at 0.3 ppm, assigned to Cl-Si-CH_3 , disappeared, and the characteristic signals observed at 0 and 3.6 ppm could be attributed to $>\text{Si}(\text{CH}_3)_2$ and $-\text{O}-\text{C}_2\text{H}_4-\text{O}-$, respectively. The ratio of the integrated intensity of the two kinds of protons changed corresponding to the change in the feed monomers. From gel permeation chromatography (GPC) measurements, the molecular weight of each DMS-*n*EO copolymer was estimated as about 3000 on the basis of narrow distribution PEO standards. Figure 1 shows the relationship between T_g and EO content represented in mole % for the DMS-*n*EO copolymers. T_g of PEO was -66°C , while that of poly(dimethyl siloxane) (PDMS) was -123°C . In the DSC curve for each DMS-*n*EO copolymer, only one endotherm corresponding to T_g was observed in the temperature range of -123 to -66°C . T_g 's of DMS-*n*EO copolymers were between -103 and -75°C , and increased with the increase in the EO content. The change in the T_g as a function of EO content was given by the Gordon and Taylor equation:¹³

$$T_{gAB} = V_A T_{gA} + V_B T_{gB}$$

where T_{gA} ($= -66^\circ\text{C}$) and T_{gB} ($= -123^\circ\text{C}$) are the T_g of PEO and PDMS, respectively,

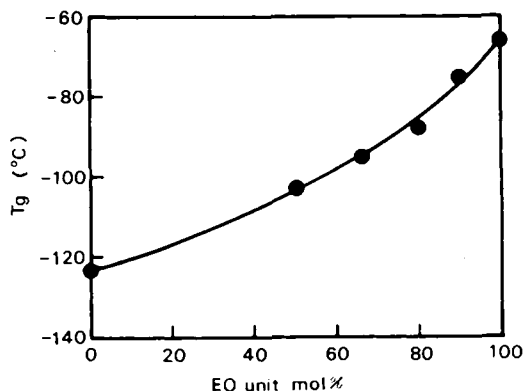


Fig. 1. Relationship between T_g and EO content for DMS-*n*EO copolymers.

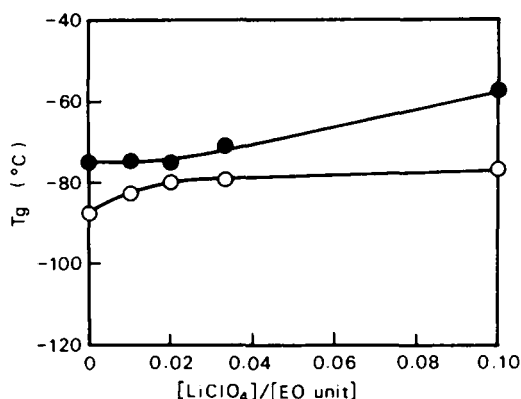


Fig. 2. Relationship between T_g and LiClO_4 concentration for DMS- n EO copolymers ($n = 4$ and 9). (○) DMS-4EO, (●) DMS-9EO.

and V_A and V_B are the volume fractions of EO and DMS units, respectively. The calculated curve (solid line in Fig. 1) fit the experimental data very well. These results showed that the copolymerization between DDS and n EG proceeded, and that the introduction of DMS unit decreased the T_g 's of the copolymers.

DMS- n EO copolymers could dissolve LiClO_4 independent of the EO content. Figure 2 shows the relationship between T_g and LiClO_4 concentration for the DMS- n EO copolymers ($n = 4, 9$). The T_g 's of both copolymers increased with the increase in LiClO_4 concentration. When the LiClO_4 concentration increased from 0 to 0.1, the increase in T_g was 10°C for DMS-4EO and 20°C for DMS-9EO. Since PDMS did not dissolve LiClO_4 , the increase in T_g seemed to be caused by the solvation of the ether oxygen atoms in EO units to LiClO_4 .^{14,15} It had been reported that the T_g of poly(propylene oxide), which has a similar structure to PEO, increased by 50°C with the increase in the LiClO_4 concentration from 0 to 0.1.¹⁴ Thus, the difference in the increase of T_g between both copolymers is due to the difference in the sequence of EO units in the copolymers. The solvation site to LiClO_4 seems to be EO units in the copolymers, and the DMS units may sterically hinder the solvation of the EO units to LiClO_4 .

Figure 3 shows the conductivity at 25°C for the DMS- n EO solutions as a function

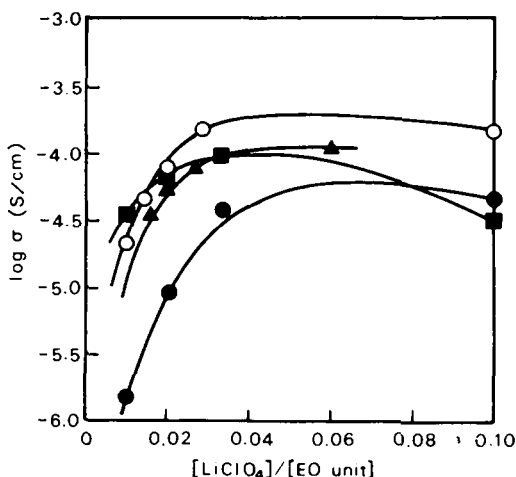


Fig. 3. Conductivity at 25°C for DMS- n EO solutions as a function of LiClO_4 concentration. (●) DMS-1EO, (▲) DMS-2EO, (○) DMS-4EO, (■) DMS-9EO.

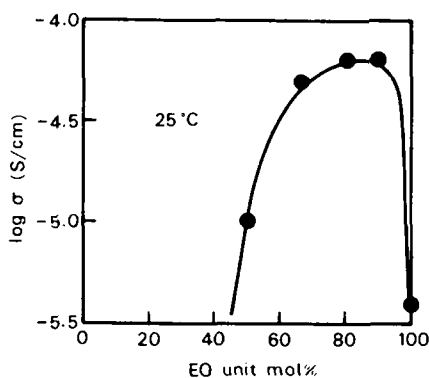


Fig. 4. Conductivity at 25°C for DMS-*n*EO solutions ($[\text{LiClO}_4]/[\text{EO unit}] = 0.02$) as a function of EO content.

of the LiClO_4 concentration. The enhanced conductivity was observed with the dissolution of LiClO_4 in the copolymers, which showed that LiClO_4 dissociated and that the conductivity was due to migration of the dissociated ions. The highest conductivity (1.5×10^{-4} S/cm) was observed in the solution of DMS-4EO copolymer at $[\text{LiClO}_4]/[\text{EO unit}] = 0.03$. It should be noted that the conductivity is comparable to that of solutions formed by the usual organic solvents with low molecular weights. It is well known that conductivity is represented by $\sigma = \sum ne\mu$, where n is the number of carriers, e is the charge of a carrier, and μ is the carrier mobility. Thus, the increasing conductivity in the low-concentration region might be attributed to the increase in the number of ionic carriers. At the higher-concentration region the conductivity became nearly constant, or, in the case of the DMS-9EO solution, decreased gradually. This behavior is due to the decrease in the carrier mobility.^{10,11} The increase in the T_g 's of the solutions with the increase in the LiClO_4 concentration depressed the segmental mobility of the polymer backbone which correlated to the carrier mobility.

Figure 4 shows the conductivity at 25°C for the DMS-*n*EO solutions ($[\text{LiClO}_4]/[\text{EO unit}] = 0.02$) as a function of EO content in the copolymers. The maximum conductivity observed at the EO content of 80–90 mol % was significantly higher than that of the PEO solution. Since the T_g 's of the solution decreased with the decrease in the EO content, the carrier mobility correlated with the segmental mobility is thought to increase with the decrease in the EO content. On the other hand, the increase in the EO content seems to accelerate the generation of ionic carriers. The appearance of the maximum conductivity can be attributed to the addition of the both effects.

From the above results it was concluded that the introduction of DMS units to PEO by copolymerization contributed to the increase in the carrier mobility; however, it suppressed the generation of ionic carriers. Thus, the highest conductivity at 25°C (1.5×10^{-4} S/cm), which was considerably higher than that of the PEO solution, was observed in the DMS-4EO solution at $[\text{LiClO}_4]/[\text{EO unit}] = 0.03$. Studies on the ion conductive behavior in these copolymer solutions and in their crosslinked films will be reported in the near future.

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